

Shape optimization of airfoils in transonic flow using a multi-objective genetic algorithm

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Abstract

Shape optimization of transonic airfoils requires creating an airfoil that reduces the drag due to transonic shocks by either eliminating them or reducing their strength at a given transonic cruise speed while maintaining the lift. The RAE 2822 and NACA 0012 airfoils are most widely used test cases for validation of computational modeling in transonic flow. This study employs a multi-objective genetic algorithm for shape optimization of RAE 2822 and NACA 0012 airfoils to achieve two objectives, namely eliminating shock and maintaining or increasing the lift at a given transonic Mach number and angle of attack. The commercially available software FLUENT is employed for calculation of the flow field using the Reynolds-averaged Navier–Stokes equations in conjunction with a two-equation turbulence model. It is shown that the multi-objective genetic algorithm can generate superior airfoils compared with the original airfoils by achieving both the objectives.

Keywords

Transonic airfoils, genetic algorithm, multi-objective optimization

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Introduction

Majority of modern commercial aircrafts operate at transonic cruise speeds. Therefore, in recent years, the shape optimization of an aircraft at transonic speeds to minimize the shock drag has become a topic of great interest in computational aerodynamics.^{1–3} An aircraft in motion at transonic speeds encounters a rapid increase in drag due to the shock waves appearing on various locations on its surface, especially on the wings. These shock waves cause flow instability and increase in drag force, which result in higher fuel consumption or reduced energy efficiency. Therefore, the goal of shape optimization is to design shock-free airfoils/wings to reduce shock drag. However, the drag reduction alone cannot be the optimization target if it can adversely affect the lift; in general, there is a tradeoff between lift and drag.⁴ Several important criteria must be met in the optimization process to achieve not only a shock-free but also an efficient airfoil that has good drag characteristics, a buffet boundary high enough to permit cruising at design lift coefficient, no unsatisfactory off-design performance, and so forth.⁵ In the last several decades, many important contributions have been made to the design and optimization of shock-free

airfoils. In the late 1950s, Morawetz^{6–8} determined that shock-free airfoils are special isolated solutions and a small perturbation from them would cause the appearance of a shock. In the 1960s, Whitcomb and Clark⁹ developed their famous supercritical airfoils for transonic flow and Garabedian and Korn¹⁰ developed the successful method of complex characteristics in the hodograph plane for obtaining shock-free airfoils. Using the hodograph method, Nieuwland¹¹ reported shock-free transonic flows around quasi-elliptical airfoils sections in 1967.

The goal of this article is to employ a multi-objective genetic algorithm (MOGA) for shape optimization of some well-known airfoils, namely the RAE 2822 and NACA 0012, at transonic speed to improve their aerodynamics characteristics, that is to reduce their drag by achieving shock-free shapes as well as to maintain or increase their lift in an

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acceptable range. The commercially available software ANSYS-FLUENT¹² is used for calculation of the flow field using the Reynolds-averaged Navier–Stokes equations in conjunction with a two-equation turbulence model. Using MOGA, globally optimal airfoils are obtained for given flow conditions and angle of attack.

Brief literature review of RAE 2822 and NACA 0012 airfoils

The goal of this study is to conduct shape optimization of two airfoils, namely the RAE 2822 and NACA 0012, using MOGA to reduce their drag as well as to maintain or increase their lift in an acceptable range. The RAE 2822 airfoil has a maximum thickness of 12.1% chord length. Here, “RAE” stands for Royal Aircraft Establishment. It is a supercritical airfoil, which has become a standard test case for turbulence modeling validation.¹³ Figures 1 and 2 show the airfoil shape and pressure contours, and pressure distribution, respectively, for RAE 2822 at $M=0.75$, $Re=6.2 \times 10^6$, and $\alpha=2.8^\circ$.

The NACA 0012 is another widely used airfoil for computational fluid dynamics (CFD) validation and shape optimization studies in transonic flow. The airfoil section shape of NACA 0012 is given by the following equation¹⁴

$$y_t = \frac{t}{0.2} c \left[0.2969 \sqrt{\frac{x}{c}} - 0.1260 \left(\frac{x}{c} \right) - 0.3516 \left(\frac{x}{c} \right)^2 + 0.2843 \left(\frac{x}{c} \right)^3 - 0.1015 \left(\frac{x}{c} \right)^4 \right] \quad (1)$$

where t is the ratio of maximum thickness to chord, which is 0.12 for NACA 0012. The airfoil shape given by equation (1) is shown in Figure 3.

This study starts with the airfoils shown in Figures 1 and 3 as baseline airfoils and employs the MOGA to optimize their shapes to obtain shock-free shapes. The MOGA results for the lift coefficient, the drag coefficient, and lift to drag ratio are compared with the values of these coefficients for the original airfoils given in previous studies.^{15–17} In all the optimization studies, the maximum thickness of the airfoils is allowed to vary within 2% of the chord from the baseline shapes.

Brief description of genetic algorithm and airfoil parameterization

This section describes in brief the single-objective genetic algorithm and MOGA techniques, followed by the parameterization of the airfoils.

Single-objective genetic algorithm

Genetic algorithms (GAs) are a class of stochastic optimization algorithms inspired by the biological evolution. In GA, a set or generation of input vectors, called individuals, is iterated over, successively combining traits (aspects) of the best individuals until a convergence is achieved. In general, GA employs the following steps.¹⁸

1. **Initialization:** randomly create N individuals.
2. **Evaluation:** evaluate the fitness of each individual.

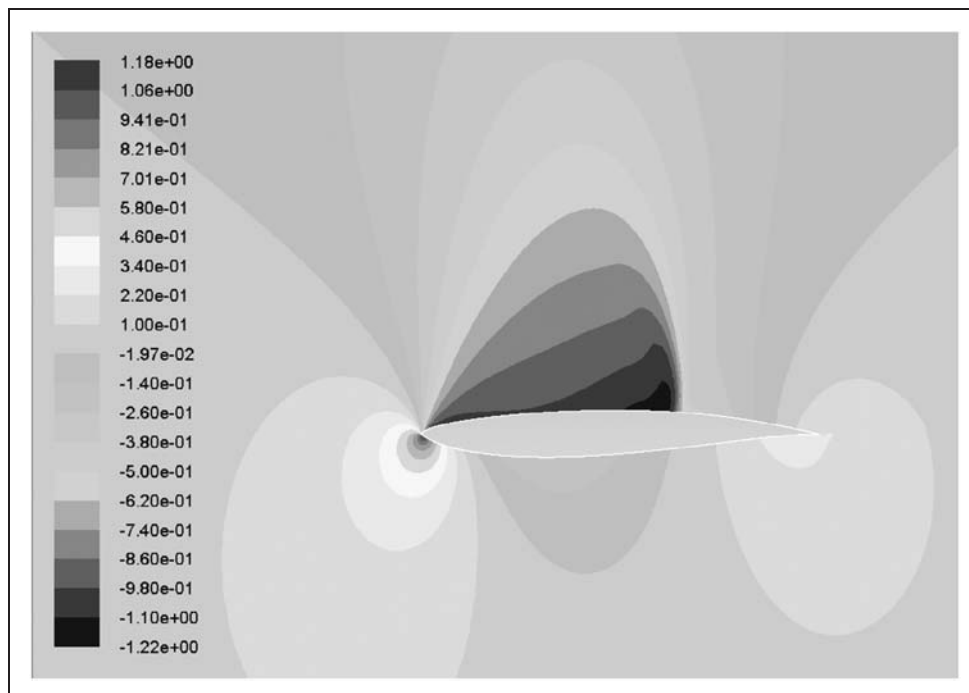


Figure 1. RAE 2822 airfoil: airfoil shape and pressure contours at $M=0.75$, $Re=6.2 \times 10^6$, $\alpha=2.8^\circ$.

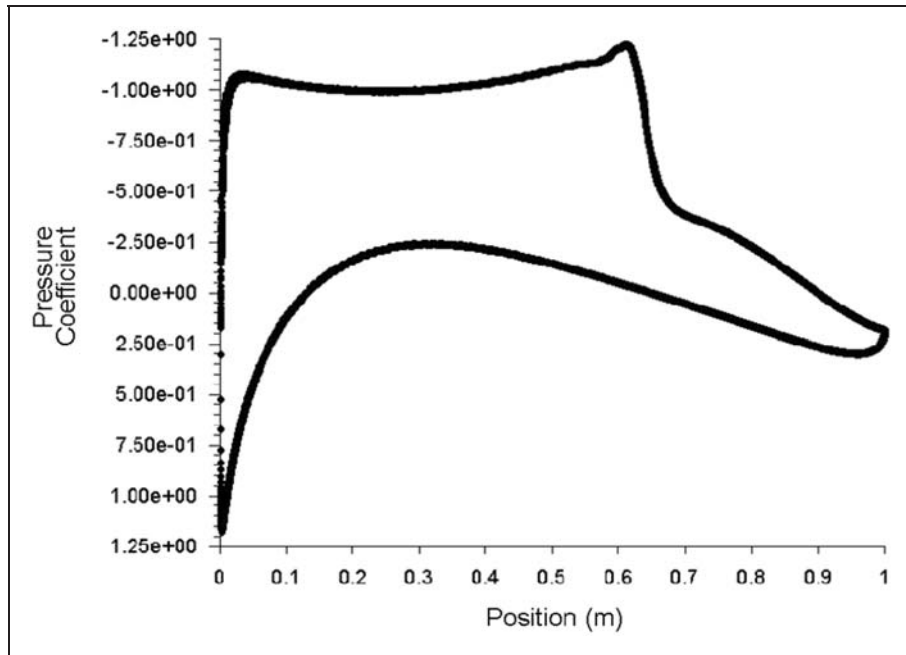


Figure 2. RAE 2822 airfoil: pressure distribution at $M = 0.75$, $Re = 6.2 \times 10^6$, $\alpha = 2.8^\circ$.

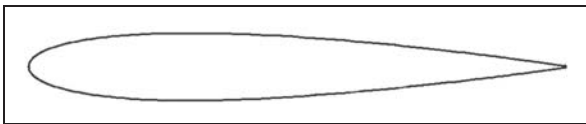


Figure 3. Geometry of the blade section of NACA 0012 airfoil.

3. **Natural selection:** remove a subset of the individuals. Often the individuals that have the lowest fitness are removed according to the removal percentage defined at the beginning of GA.
4. **Reproduction:** pick pairs of individuals to produce an offspring. This is often done by roulette wheel sampling, that is the probability of selecting some individual h_i for reproduction is given by¹⁸

$$P[h_i] = \frac{\text{fitness}(h_i)}{\sum_j \text{fitness}(h_j)} \quad (2)$$

5. A crossover function is then performed to produce the offspring. Generally, crossover is implemented by choosing a crossover point on each individual and swapping alleles or vector elements at this point, as illustrated in Figure 4.
6. **Mutation:** randomly alter some small percentage of the population.
7. **Check for convergence:** if the solution has converged, return the best individual observed. If the solution has not yet converged, label the new generation as the current generation and go to step 2. Convergence is often defined by a certain number of generations or a similarity threshold.

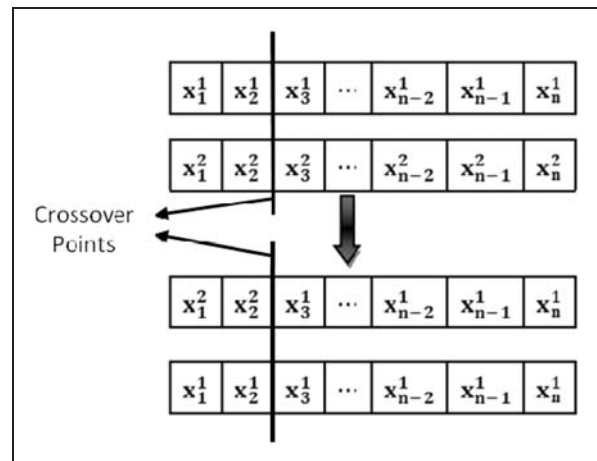


Figure 4. Illustration of the crossover function in GA: choosing crossover point and swapping alleles. GA: genetic algorithm.

Multi-objective GA

For many design problems, it is desirable to achieve, if possible, simultaneous optimization of multiple objectives.¹⁹ These objectives, however, are usually conflicting, preventing simultaneous optimization of each objective.²⁰ Therefore, instead of searching for a single optimal solution, a MOGA is necessary to find a set of optimal solutions (generally known as Pareto-optimal solutions). In this study, the MOGA algorithm is used to find the Pareto-optimal solutions to the airfoil optimization problem. The widely known Non-dominated Sorting Genetic Algorithm II (NSGA-II) is employed.²¹

In NSGA-II, a Pareto-optimal solution set is defined such that any individual inside the set dominates any individual outside the set while any individual in the set is not dominated by another individual in the solution set. Here, a feasible solution $\mathbf{x}$ is said to dominate another feasible solution $\mathbf{y}$, if and only if all the objectives of $\mathbf{x}$ are not worse than those of $\mathbf{y}$ and at least one objective of $\mathbf{x}$ is better than that of $\mathbf{y}$. A solution is said to be Pareto-optimal if it is not dominated by any other solution in the solution space.²⁰ The NSGA-II used in this study has the following three features: (a) it uses an elitist principle; (b) it uses an explicit diversity preserving mechanism; and (c) it emphasizes non-dominated solutions in a population.²² The implementation procedure of NSGA-II is as follows²¹

1. At **0-th generation**, a random parent population P_0 of size N is created; it is sorted based on the non-domination. Then the individuals in P_0 are ranked: 1 is the best level, 2 is the next-best level, and so on. Then P_0 is sent to selection, cross-over, and mutation operators to create offspring population Q_0 of size N .
2. At **t -th generation**, a combined population $R_t = P_t \cup Q_t$ of size $2N$ is formed and is sorted according to non-domination. Then individuals in R_t are divided into the best non-dominated set F_1 , the next-best non-dominated set F_2 and so on. If the size of F_1 is smaller than N , all members of F_1 go to P_{t+1} , with the remaining members chosen from $F_2, F_3, \dots$ until the size of P_{t+1} is N . Then new population P_{t+1} is sent to selection, crossover, and mutation operators to create a new population Q_{t+1} of size N .
3. **Termination:** the procedure terminates when convergence criterion is met.

This study utilizes a java code package called jMetal. It is a Java-based framework for multi-objective optimization using meta-heuristics. It is easy-to-use and is flexible and extensible.²³

Airfoil parameterization

In all the optimization studies considered in this article, the maximum thickness is allowed to vary within 2% of the chord of the baseline airfoils. Table 1 shows the maximum thickness constraints for the two airfoils.

The airfoil shapes are parameterized using Bezier curves. Bezier curves are parametric curves frequently used in computer graphics and relate fields. A Bezier curve is defined by a set of Bezier control points. Each curve can be expressed as math equations containing the information of Bezier control points. The number of points required to parameterize a curve depends on the shape of the curve.

Figure 5 is an example of a Bezier curve and Bezier control points for one airfoil. Each airfoil is divided

Table 1. Maximum thickness constraints used for airfoil shape optimization in MOGA.

	RAE 2822	NACA 0012
Lower bound	0.11	0.11
Upper bound	0.13	0.13
Baseline	0.121	0.12

MOGA: multi-objective genetic algorithm.

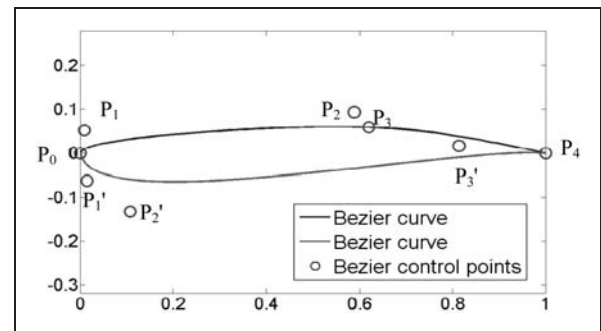


Figure 5. Example of Bezier curves and Bezier control points.

into top and bottom boundary curves by the chord, shown in different colors in Figure 5. Since the GA optimization involves fitness evaluation for a large number of airfoils with various shapes, five control points are used for both the top and bottom boundaries, thus the degree of the Bezier curve is four. For an airfoil curve, the first P_0 and the last point P_4 are fixed since they represent the leading and trailing edge of the airfoil. The intermediate points are allowed to move within the boundaries. At the top surface, three control points $P_1 (x_1, y_1)$, $P_2 (x_2, y_2)$, and $P_3 (x_3, y_3)$ are used. At the bottom surface, three control points $P_1' (m_1, n_1)$, $P_2' (m_2, n_2)$, and $P_3' (m_3, n_3)$ are employed. The constraints applied to the Bezier control points are shown in Table 2.

Algorithm implementation

This section describes the computational setup. Figure 6 schematically illustrates how the MOGA interfaces with the external mesh generator “GAMBIT”²⁴ and CFD solver “FLUENT.”¹²

Implementation of MOGA

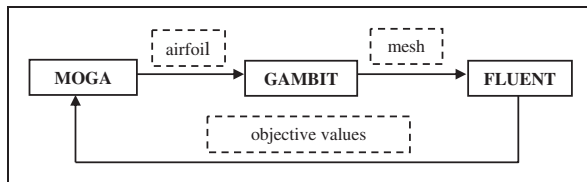
The implementation of MOGA involves the following steps.

1. **Initial generation creation.** At the initial generation, the MOGA code generates a number of individuals according to the generation size. Since each individual represents one feasible airfoil, it contains the coordinates of the Bezier

Table 2. Bezier control points and their lower/upper bounds used for airfoil shape optimization in MOGA.

Airfoil	Bezier points coordinates	RAE 2822		NACA 0012	
		Lower bound	Upper bound	Lower bound	Upper bound
Top surface	x_1	0	0.03	0	0.03
	x_2	0.03	0.6	0.03	0.8
	x_3	0.6	0.9	0.8	1
	y_1	0.02	0.06	0.02	0.09
	y_2	0.03	0.11	0.04	0.11
	y_3	0.04	0.11	0.04	0.11
Bottom surface	m_1	0	0.04	0	0.04
	m_2	0.04	0.45	0.04	0.4
	m_3	0.45	0.9	0.4	1
	n_1	-0.06	-0.02	-0.09	-0.02
	n_2	-0.2	-0.02	-0.2	-0.02
	n_3	-0.02	0.02	-0.04	0.02

MOGA: multi-objective genetic algorithm.

**Figure 6.** Illustration of information flow in MOGA process. MOGA: multi-objective genetic algorithm.

control points, which form the airfoil shapes using Bezier curves.

2. **Mesh creation.** Each individual is converted into real airfoil shape using the Bezier equations. Then their coordinates are written in a data file, which is passed on to the meshing program “GAMBIT.”²⁴ GAMBIT is used to create a two-dimensional structured or unstructured mesh. It first reads a previously written journal file and makes the mesh following the instructions from the journal file.
3. **Flow-field computation.** When a mesh is generated, it is then passed as input to the CFD flow solver FLUENT¹² for computation of the flow field using a journal file. FLUENT is used iteratively to solve for the coefficient of lift C_l and the coefficient of drag C_d . Some combination of these values (either C_l and/or C_l/C_d) is taken as the quantities of interest (single or multiple objective values) to determine the objectives of the individual airfoil.
1. **Optimization and new generation creation.** When the objectives of all the individuals in the current generation are computed, the MOGA code collects them and uses them to determine the quality

of each individual. After going through all the procedures for GA optimization described in the previous section, a new generation is created and all the steps of the GA are repeated for the new generation. The MOGA algorithm continues until the convergence in the objective value/s is achieved.

Employing the NSGA-II and the jMetal multi-objective GA software package mentioned in the previous section, this study employs a generation size of 20 individuals with the total number of generations of approximately 100. The crossover rate is 0.9. The mutation rate is determined by $1/(\text{variable number})$. There are a number of operators included in the jMetal framework for crossover, mutation, and selection. In this study, the crossover section uses the simulated binary crossover operator. The mutations section uses the polynomial mutation operator. The selection section uses the binary tournament operator.

There are two objectives in the calculation: objective 1 is to minimize $(100 \times C_d)$ and objective 2 is to minimize $(1000 \times C_d/C_l)$. Figure 7 shows an example of the Pareto solutions of the optimized RAE2822 airfoil. The x -axis is objective 1 and the y -axis is objective 2; they are both normalized to unity. Since this study is more interested in C_d (objective 1), the leftmost point (with the minimum of objective 1 in the Pareto front) is chosen as the solution to the optimization problem. Then the convergence is determined in the following way: when the leftmost point in the Pareto front of each generation stops moving for a number of generations, which means the solution with the best performance of C_d after each generation is not getting better, it is considered that the convergence criteria is met. Usually the convergence happens in 100 generations or less.

Mesh generation

The commercially available software “GAMBIT”²⁴ is used to generate a structured C-mesh for both the airfoils. Considering that NACA 0012 airfoil has a finite thickness at the trailing edge, a different shape of the domain is employed compared with that for the RAE 2822 airfoil, which has a zero thickness trailing edge. A journal file is used to automatically produce a mesh that FLUENT¹² can use to evaluate the objective functions for an airfoil. The journal must be robust enough to create a usable mesh around any arbitrary airfoil. Therefore, the far-field boundary geometry of every mesh is held identical, but the airfoil in the center of the mesh changes every time. The domain size of the mesh for RAE 2822 airfoil is such that both the radius of the front circle part and the half-width of the rectangle part are $20c$, and the length of the rectangle part is $50c$. The domain size of the mesh for NACA 0012 airfoil is between $15c$ and $20c$. Faces of the grid are meshed using quadrilateral cells,

and the numbers of nodes on opposite faces are required to be identical. To ensure this distribution, a set number of nodes (and not relative node spacing) is defined along each edge. Otherwise, thicker or more cambered airfoil edges would have more nodes than thinner ones if a relative distribution was used. Figures 8 and 9 show typical C-mesh around the airfoil, respectively. The interior of the boundary is meshed with a quadrilateral cell scheme. The resulting grid for Figure 8 has a node/cell/face count of 141,300/140,000/281,300 and that for Figure 9 has a node/cell/face count of 40,950/40,500/81,450. When meshing is completed, the boundaries are defined and a 2D mesh is written to the file.

Flow field computation

FLUENT¹² is used to solve for the coefficients of lift and drag for a given airfoil. A journal file is used to automatically initialize and evaluate each airfoil while saving a record of the relevant coefficients, C_l and C_d .

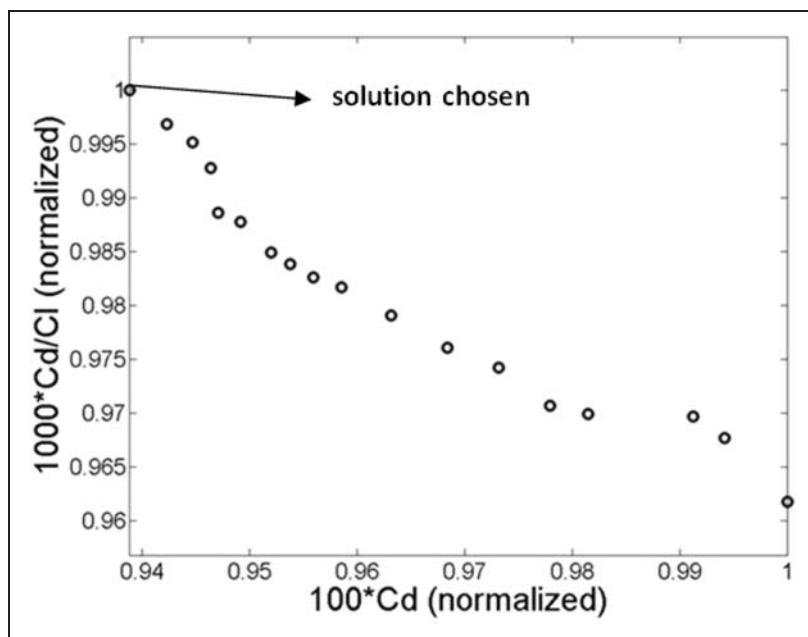


Figure 7. Pareto solutions for RAE 2822 airfoil.

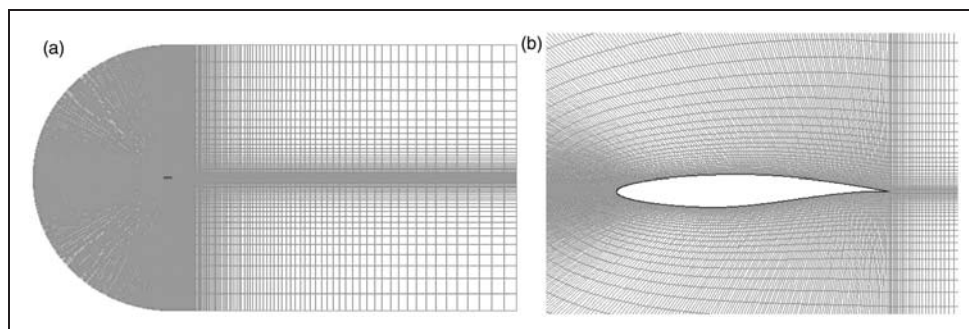


Figure 8. C-grid around the RAE 2822 airfoil. (a) Entire grid (b) Zoomed-in-view of the grid.

The FLUENT journal initializes the calculations for a specific Mach number for each case. In this study, a two-equation $k-\omega$ SST Turbulence Model with low-Re corrections and compressibility effects is employed. The solver type is density-based and the solution method is implicit. The air is considered as an ideal gas and the laminar viscosity is calculated from the Sutherland's law

$$\mu = \mu_o \left(\frac{T}{T_o} \right)^{3/2} \left(\frac{T_o + S}{T + S} \right)$$

with $\mu_o = 1.716 \times 10^{-5} \text{ kg/m s}$, $T_o = 273 \text{ K}$, and $S = 111 \text{ K}$

Various Mach numbers are used in the flow field calculation. For the cases with Mach number of 0.73 and 0.75, the temperature is defined to be at the room temperature (300 K) considering that the experimental data came from the wind tunnel tests; the static pressure is 101,325 Pa. For the case of Mach number of 0.8, where the compressibility effects dominate the flow field near the airfoil, the temperature is 275.709 K and the static pressure is 66470 Pa according to the compressible flow equations

$$\frac{p}{p_o} = \left[1 + \left(\frac{\gamma - 1}{2} \right) M^2 \right]^{-\frac{\gamma}{\gamma - 1}}$$

where p_o = total pressure = 101,325 Pa and $\gamma = 1.4$ for air

$$\frac{T}{T_o} = 1 + \left(\frac{\gamma - 1}{2} \right) M^2$$

where T_o = total temperature = 311 K and $\gamma = 1.4$ for air

Results and discussion

Optimization results

In this study, our primary objective has been to generate optimized NACA 0012 and RAE 2822 airfoils that minimize the shock strength in transonic flow without reducing the lift. Four flow cases for the original NACA 0012 and RAE 2822 airfoils are used as baseline cases for optimization. Table 3 shows the flow conditions, angles of attack, wind tunnel experimental results, and computational validation results using FLUENT. Before performing the optimization, CFD solutions were validated using the wind tunnel data for the baseline airfoils. The same mesh and computational settings were then applied to the new optimized airfoils created by the MOGA process. The wind tunnel data for RAE 2822 I airfoil was obtained from the study of Padovan et al.¹³ while the data for RAE 2822 II was obtained from Abdol-Hamid et al.¹⁴

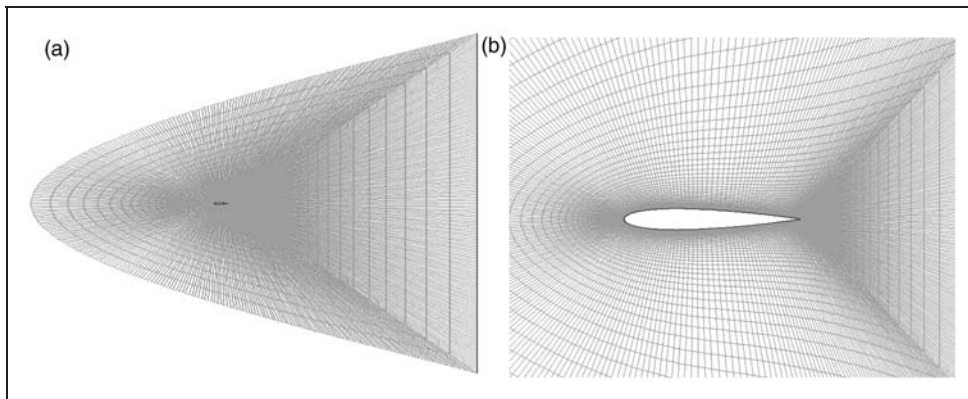


Figure 9. C-grid around the NACA 0012 airfoil. (a) Entire grid (b) Zoomed-in-view of the grid.

Table 3. Baseline airfoils test cases with experimental and computational results.

Airfoil	Mach number	Angle of attack	C_l/C_d		C_l		C_d	
			Test	CFD	Test	CFD	Test	CFD
RAE 2822 I	0.73	2.0	39.41	54.17	0.67	0.65	0.017	0.012
RAE 2822 II	0.75	2.8	30.7	28.9	0.743	0.7387	0.0242	0.02556
NACA0012 I	0.75	2.0	19.29	19.34	0.328	0.3206	0.017	0.01658
NACA0012 II	0.80	1.5	7.16	6.18	0.265	0.26539	0.037	0.04294

CFD: computational fluid dynamics.

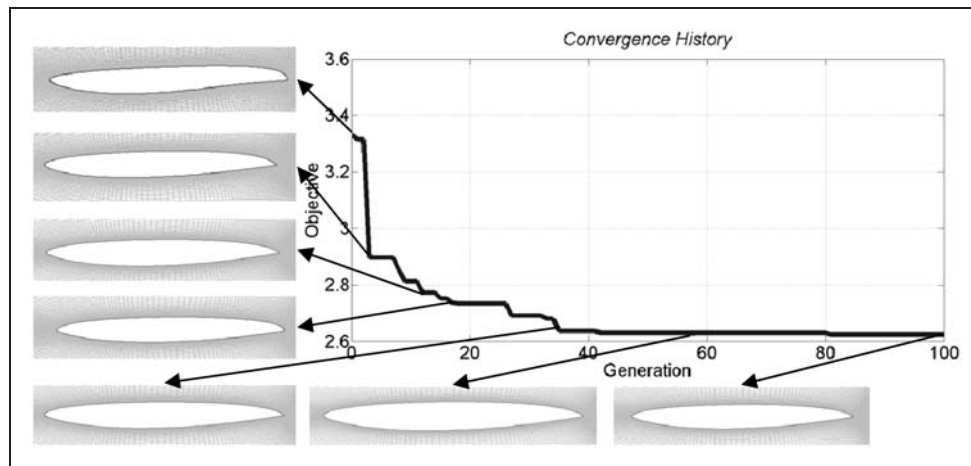


Figure 10. Convergence history of NACA 0012 II optimized airfoil using the genetic algorithm.

Table 4. Aerodynamic properties of airfoils at different generations.

Generation number	Objective 1 $100 \times C_d$	C_l	C_d	C_l/C_d
0	3.331022	0.24673	0.03331	7.407023
3	2.896753	0.289217	0.028968	9.984172
9	2.812022	0.279371	0.02812	9.93489
18	2.733454	0.265976	0.027335	9.730389
36	2.63671	0.267701	0.026367	10.15285
58	2.629238	0.26618	0.026292	10.12385
100	2.622685	0.26522	0.026227	10.11248

The experimental data for NACA0012 cases I and II was obtained from the study of Gregory and Wilby.¹⁵

Figure 10 shows the typical “evolution” of an optimized airfoil using the genetic algorithm. This figure plots the best objective 1 ($100 \times C_d$) in each generation against the generation number. It illustrates the convergence history for NACA 0012 II case. It should be noted that the solution is nearly optimal after about 80 generations. In Table 4, the C_l , C_d and C_l/C_d of various airfoils with the best objective 1 at different generations are shown. They converge to a value of $C_l=0.26522$, $C_d=0.026227$ and $C_l/C_d=10.11$ after about 80 generations. Generally speaking, the total computational time for all the cases is about 40 h, depending on the number of parallel processors in the computing platform. This is quite efficient considering the number of generations needed to obtain the optimized airfoils.

Table 5 shows the comparison of results between the original airfoils (Figures 1 and 3) and the optimized airfoils at various Mach numbers and angles-of-attack. For all the four test cases, the optimized airfoils have larger C_l/C_d and smaller C_d ; C_l is either slightly reduced or increased. The maximum thickness is also close to that of the original airfoils ($\sim 8\%$).

Figures 11–14 shows (a) the optimized shape, (b) the pressure distribution, and (c) the velocity contours for the optimized RAE 2822 and NACA 0012 airfoils at various Mach numbers and angles of attack, respectively. All the four test cases have achieved shock-free airfoils. These results clearly demonstrate the superior performance of the optimized airfoils obtained using MOGA.

Discussion

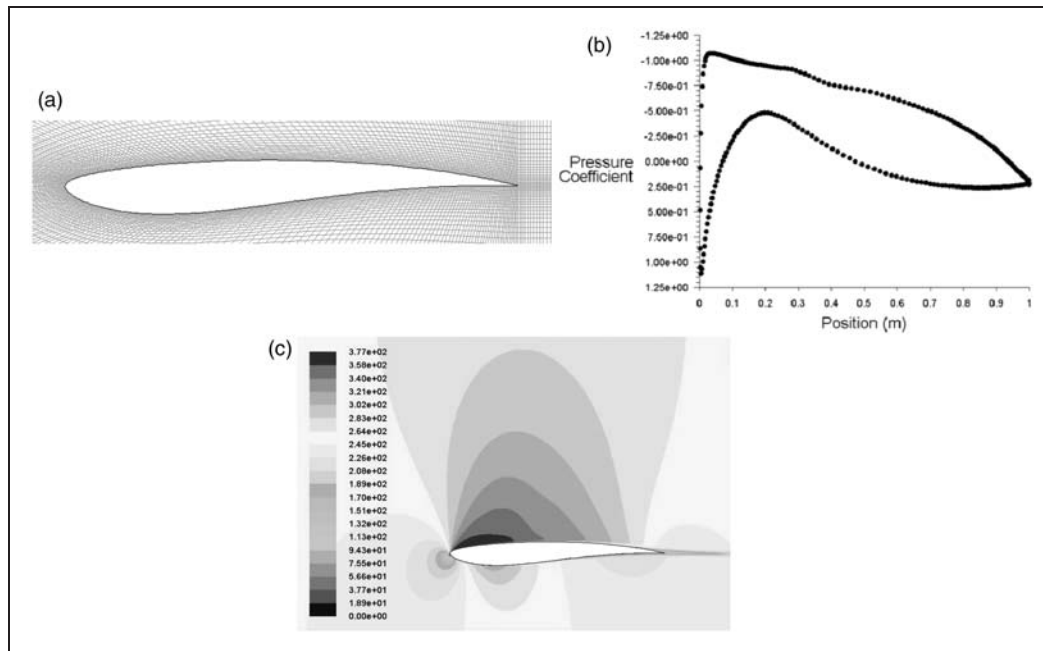
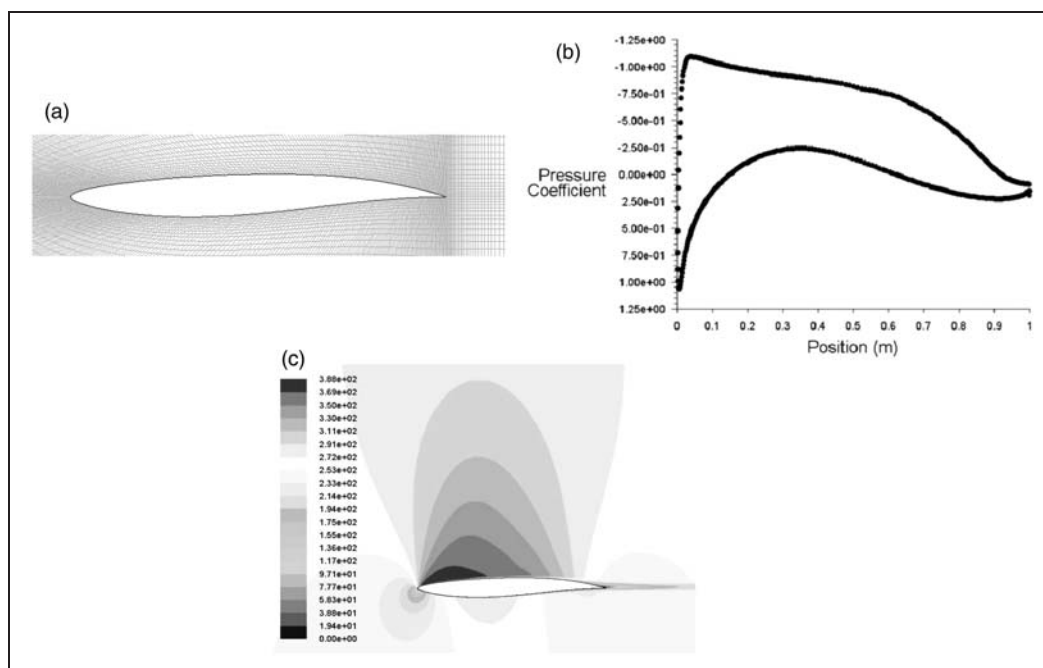
Figures 15–22 show the comparison of optimized and original airfoil shapes, pressure distributions, and pressure contours for RAE 2822 and NACA 0012 airfoils at different flow conditions obtained using MOGA.

Compared with the original airfoils, the optimized airfoils have superior performance in reducing shock drag. From the figures comparing the pressure distributions and pressure contours, one can observe that the shock waves on the top surfaces of the original airfoils are almost eliminated in the optimized airfoils. Furthermore, it can be observed that in order to maintain C_l after the elimination of shock, in all the optimized airfoils, the pressure difference between the top and bottom surface becomes greater near the leading edge. Table 6 shows that the drag coefficients of all the optimized airfoils are reduced by nearly 30–40% compared with the baseline airfoils while their maximum thicknesses are altered by less than 10%. Their lift coefficients either increase or remain the same for the optimized NACA 0012 airfoils and decrease by only about 3% for both the RAE 2822 test cases, which is within the acceptable range. All the optimized airfoils have much improved C_l/C_d increasing from 40% to 70% compared with the baseline airfoils.

The reason for a small reduction in C_l for RAE airfoils is as follows. The original RAE 2822 airfoil is already a supercritical airfoil. It had been improved

Table 5. Comparison of results for original and optimized RAE 2822 and NACA 0012 airfoils.

Airfoil	Mach number	Angle of attack	Optimized maximum thickness	C_l/C_d		C_l		C_d	
				Optimized	Original	Optimized	Original	Optimized	Original
RAE 2822 I	0.73	2.0	0.11c	57	39.41	0.65	0.67	0.0114	0.017
RAE 2822 II	0.75	2.8	0.11c	48.32	30.7	0.72	0.743	0.0149	0.0242
NACA0012 I	0.75	2.0	0.112c	33.425	19.29	0.3767	0.328	0.01127	0.017
NACA0012 II	0.8	1.5	0.11c	10.11	7.16	0.2652	0.265	0.02622	0.037

**Figure 11.** Results for optimized RAE 2822 airfoil: $M = 0.73$, $\alpha = 2^\circ$. (a) Airfoil Shape (b) Pressure Distribution (c) Velocity Contours (zoomed-in view).**Figure 12.** Results for optimized RAE 2822 airfoil: $M = 0.75$, $\alpha = 2.8^\circ$. (a) Airfoil Shape (b) Pressure Distribution (c) Velocity Contours (zoomed-in view).

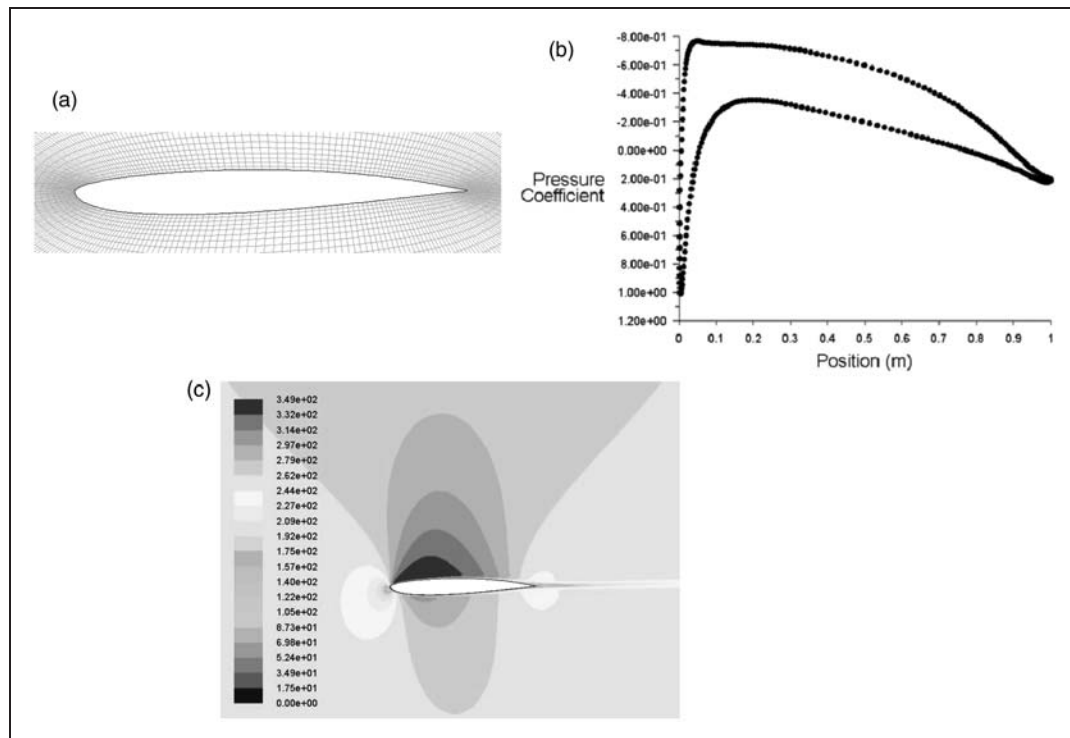


Figure 13. Results for optimized NACA 0012 airfoil: $M = 0.75$, $\alpha = 2^\circ$. (a) Airfoil Shape (b) Pressure Distribution (c) Velocity Contours (zoomed-in view).

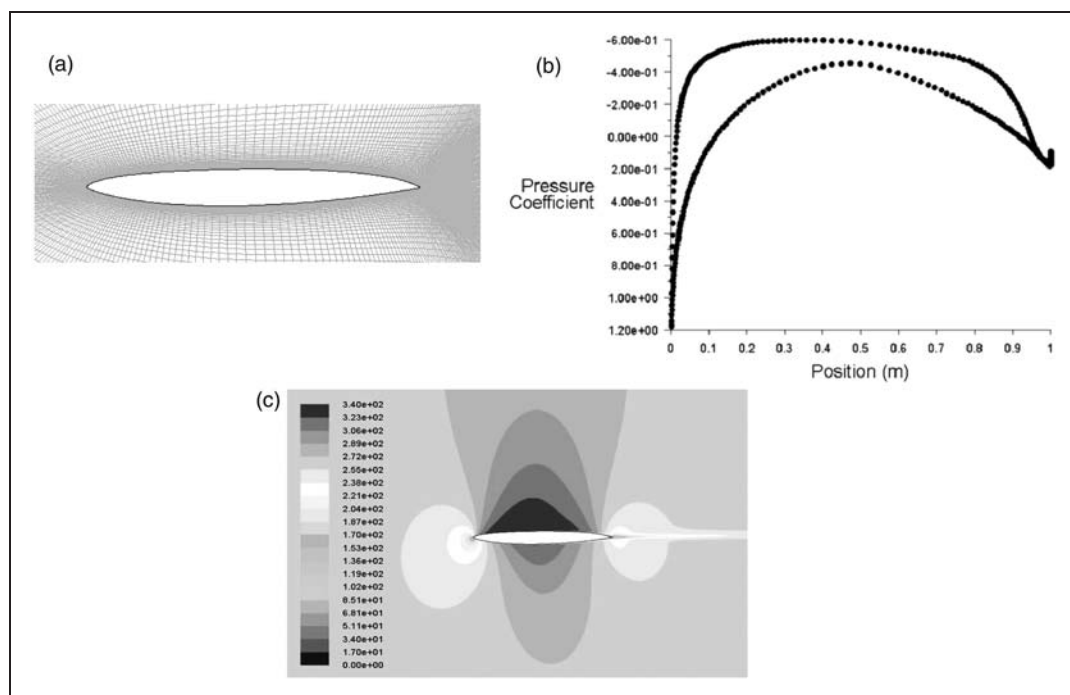


Figure 14. Results for optimized NACA 0012 airfoil: $M = 0.80$, $\alpha = 1.5^\circ$. (a) Airfoil shape (b) Pressure distribution (c) Velocity Contours (zoomed-in view).

before based on the conventional airfoil so that the supercritical airfoil shape can minimize the strength of the shock wave and create the shock wave as far as possible downstream, thus reducing drag in the transonic speed range. In order to further improve the airfoil shape so that the shock is completely

eliminated, a small reduction in the maximum thickness of the airfoil within the acceptable range is needed, which may consequently reduce the lift by a small amount. In this study, the lift reduction and shape change are both within the acceptable range.

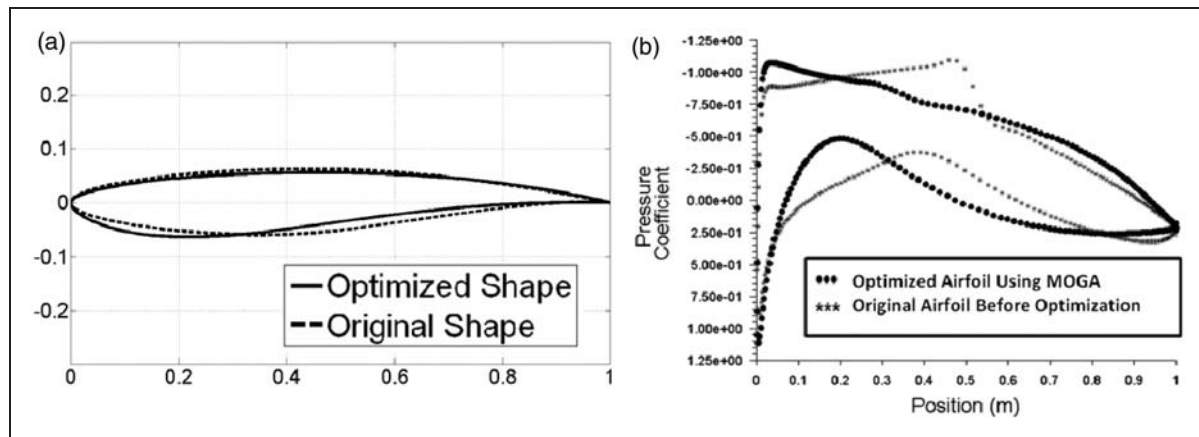


Figure 15. Comparison of original and optimized RAE 2822 airfoil shapes and pressure distributions: $M = 0.73$, $\alpha = 2^\circ$. (a) Airfoil shapes comparison (b) Pressure distributions comparison.

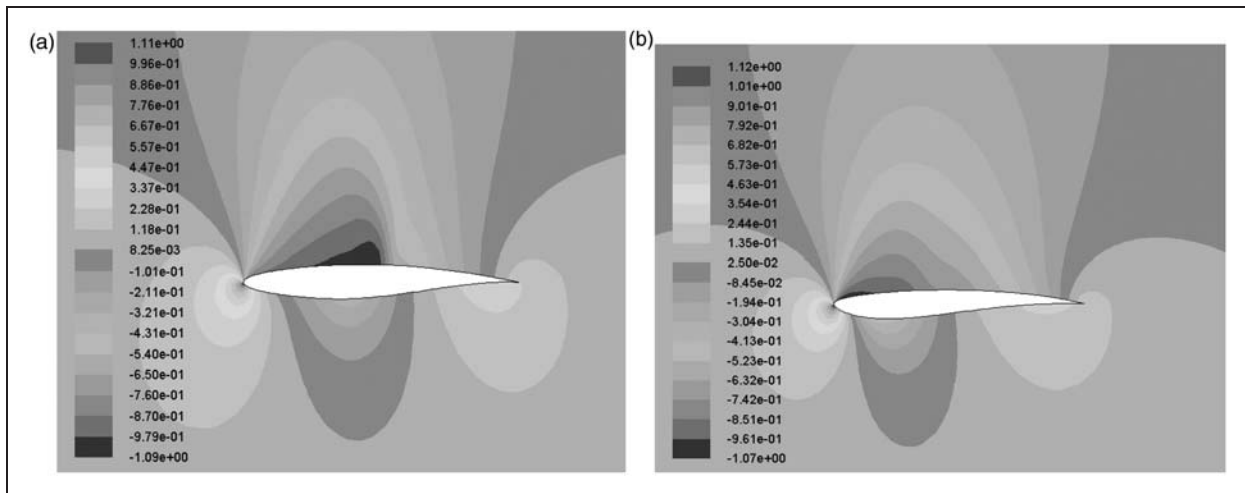


Figure 16. Comparison of original and optimized RAE 2822 airfoil pressure contours: $M = 0.73$, $\alpha = 2^\circ$. (a) Original RAE 2822 airfoil (b) Optimized airfoil.

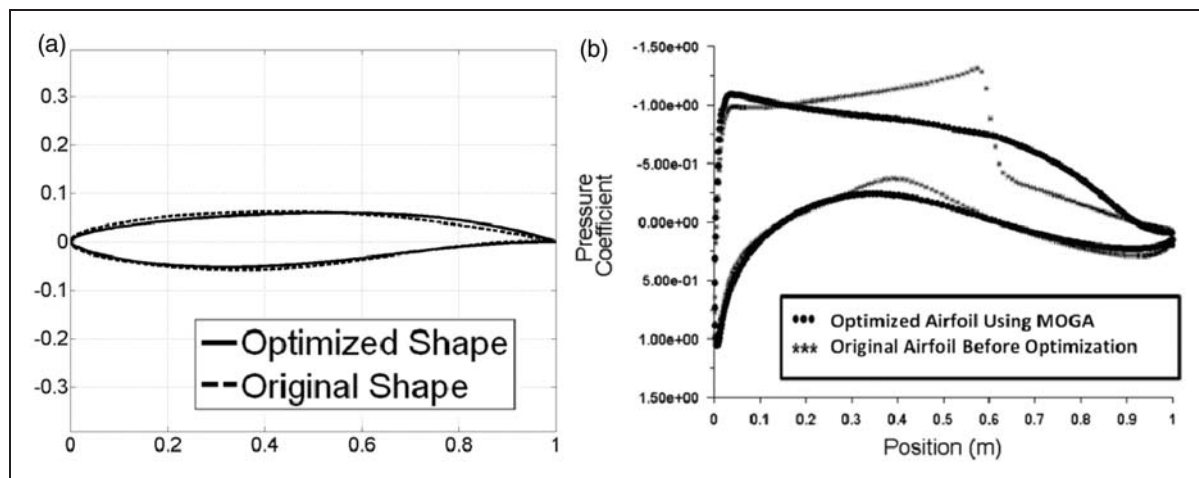


Figure 17. Comparison of original and optimized RAE 2822 airfoil shapes and pressure distributions: $M = 0.75$, $\alpha = 2.8^\circ$. (a) Airfoil shapes comparison (b) Pressure distributions comparison.

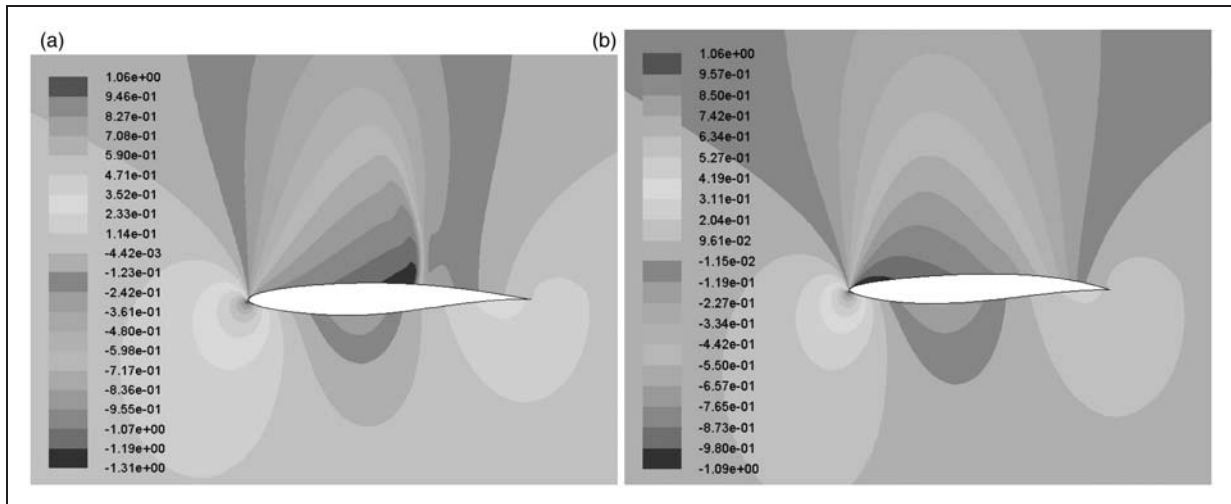


Figure 18. Comparison of original and optimized RAE 2822 airfoil pressure contours: $M = 0.75$, $\alpha = 2.8^\circ$.

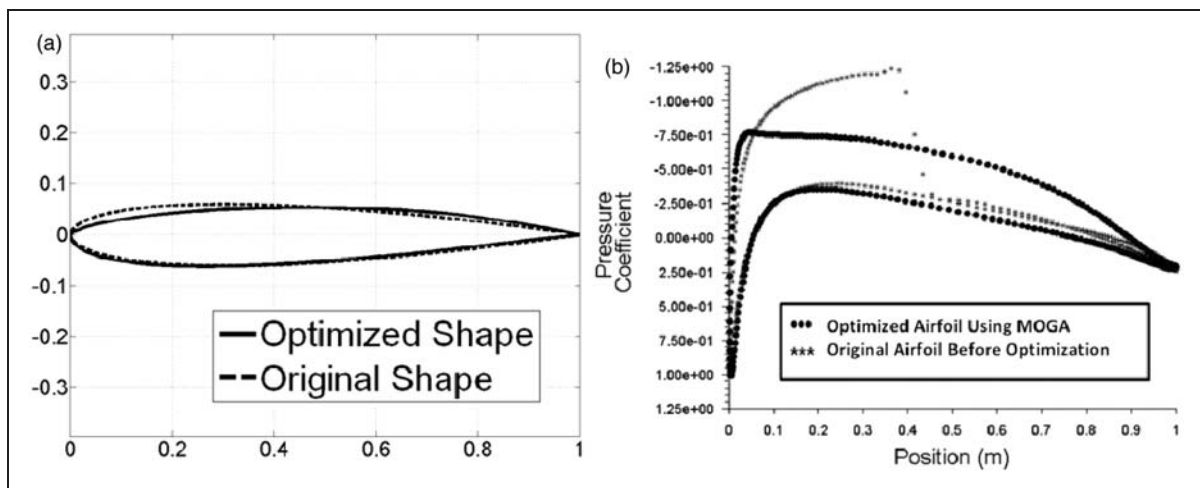


Figure 19. Comparison of original and optimized NACA 0012 airfoil shapes and pressure distribution: $M = 0.75$, $\alpha = 2^\circ$. (a) Airfoil shapes comparison (b) Pressure distributions comparison.

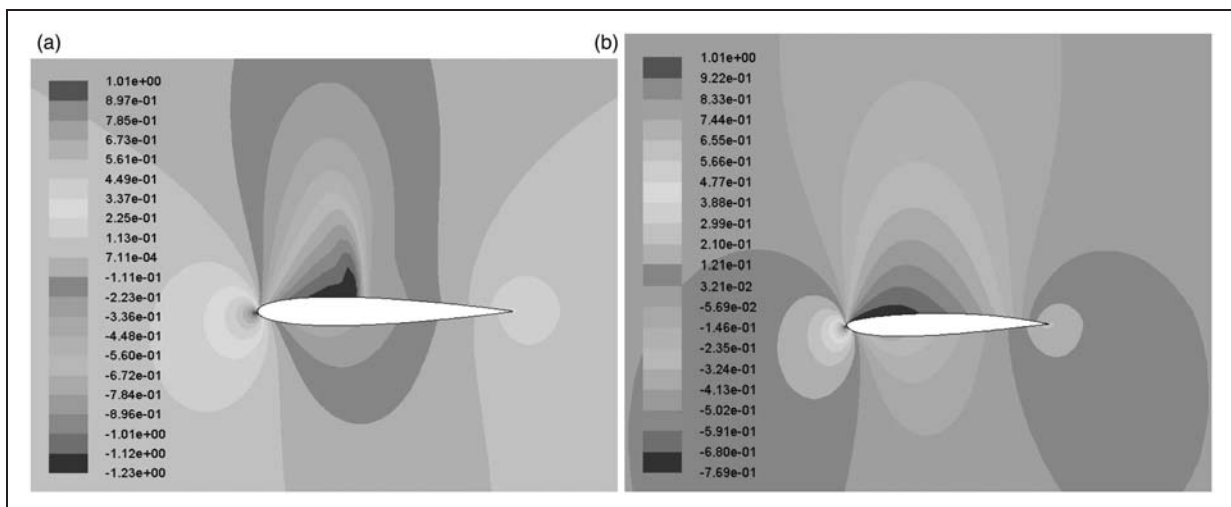


Figure 20. Comparison of original and optimized NACA 0012 pressure contours: $M = 0.75$, $\alpha = 2^\circ$.

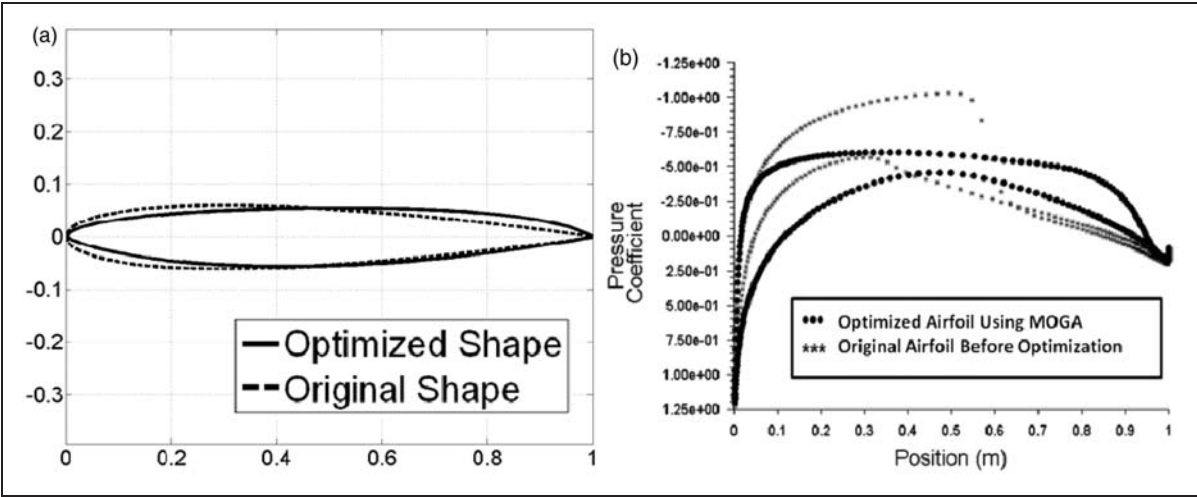


Figure 21. Comparison of original and optimized NACA 0012 airfoil shapes and pressure distributions: $M = 0.8$, $\alpha = 1.5^\circ$. (a) Airfoil shapes comparison (b) Pressure distributions comparison.

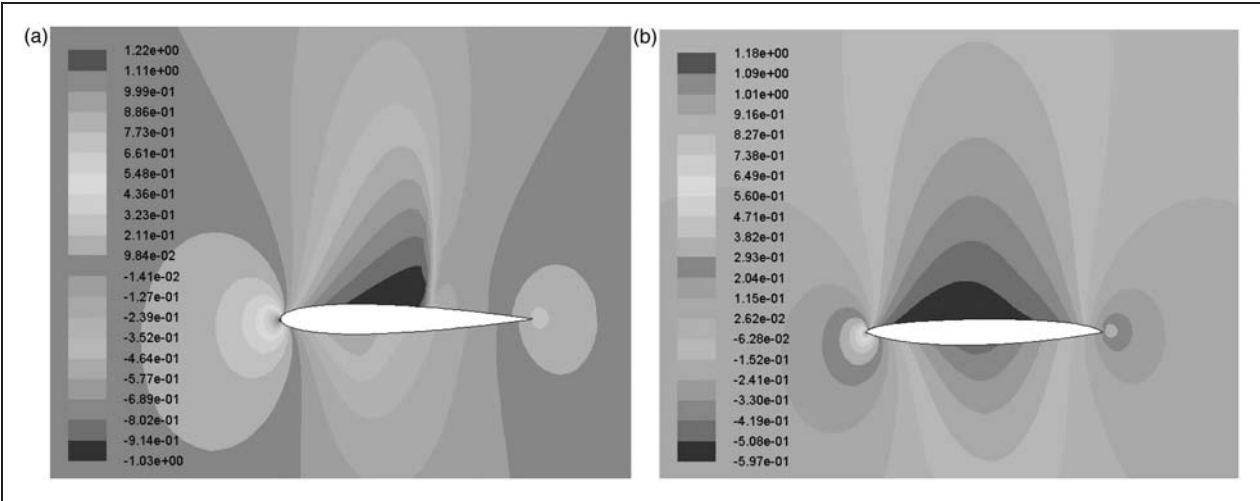


Figure 22. Comparison of original and optimized NACA 0012 airfoil pressure contours: $M = 0.80$, $\alpha = 1.5^\circ$.

Table 6. Relative improvement in the properties of optimized airfoils compared with the original airfoils.

Optimized airfoil	Change in maximum thickness (%)	Change in C_l/C_d (%)	Change in C_l (%)	Change in C_d (%)
RAE 2822 I	-9	44.6333	-2.985	-32.941
RAE 2822 II	-9	57.3941	-3.096	-38.43
NACA0012 I	7.8	73.2763	14.8476	-33.706
NACA0012 II	8.3	41.2011	0.0755	-29.135

Conclusions and recommendations

This study has made several important contributions to the study of airfoil optimization. It is shown that the GA technique can be employed efficiently and accurately to produce globally optimal airfoils with

excellent aerodynamic properties using a desired objective value (C_d and/or C_l/C_d). In addition, this study has demonstrated that the application of multi-objective GA can result in superior shock-free airfoils by minimizing the drag and maximizing the lift to drag ratio.

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Appendix

Notation

c	chord length of an airfoil
C_d	coefficient of drag
C_l	coefficient of lift
D	drag
L	lift
M	Mach number
Re	Reynolds number
t	maximum thickness of the airfoil
α	angle of attack
ρ	air density
μ	dynamic viscosity of air